

# Curriculum Vitae

Mohammad Ful Hossain Seikh [fulhossain@ku.edu](mailto:fulhossain@ku.edu)

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Current: 1607 W 9th Street Apt 6B, Lawrence, KS, USA, 66044 | +1 (785)727-6913  
Permanent: Jinarapara, Saktipur, Murshidabad, WB, India, 742163 | +91 7060617358

## Professional Objective

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To apply my training in experimental physics, radio-frequency engineering, signal processing, and data-driven analysis to research and development roles in academia, national laboratories, and industry, with a focus on instrumentation, detector systems, and advanced measurement technologies.

## Education

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| <b>Ph.D. in Physics</b> <i>2020–2026</i><br>University of Kansas, USA                     | <b>3.964/4.0</b>                    |
| <b>M.S. in Electrical Engineering</b> <i>2024–2025</i><br>University of Kansas, USA       | <b>3.778/4.0</b>                    |
| <b>M.Sc. in Physics</b> <i>2018–2020</i><br>Aligarh Muslim University, India              | <b>9.170/10</b> (Gold Medalist)     |
| <b>B.Sc. Honors in Physics</b> <i>2015–2018</i><br>Aligarh Muslim University, India       | <b>9.186/10</b> (Gold Medalist)     |
| <b>Higher Secondary (12th)</b> <i>2014</i><br>Central Board of Secondary Education, India | <b>9.120/10</b> (First Position)    |
| <b>Secondary (10th)</b> <i>2012</i><br>Central Board of Secondary Education, India        | <b>10/10</b> (Certificate of Merit) |

## Research Experience

### Selected Dissertations, Theses, and Projects

- **Physics PhD Dissertation** *2020–2026*  
Title: “Toward a Five-Station Diffuse Ultra-High Energy Neutrino Search with the Askaryan Radio Array: ARA01 Analysis Framework”. Defended on May 8, 2026. Advised by Prof. D. Besson.
- **Electrical Engineering Master’s Project** *2024–2025*  
Developed **AAFIYA: Antenna Analysis in Frequency-domain for Impedance and Yield Assessment**, a modular Python toolkit for rigorous, reproducible analysis and visualization of antenna S-parameters, impedance, and gain/beam patterns from both measurement and simulation data, with Prof. Jim Stiles at EECS Department, KU, (Report), (Slides).
- **Physics Master’s Thesis** *2019–2020*  
Investigated light sterile neutrino oscillations with Prof. M.S. Athar at Aligarh Muslim University (AMU), producing review report “Light Sterile Neutrinos: Status and Prospects” (DOI:10.13140/RG.2.2.13777.66405).
- **Physics Undergraduate Project** *2017–2018*  
“Quarks & Leptons”: Detailed study of Standard Model interactions with Prof. M.S. Athar at AMU [Handwritten Report].

## Research Roles and Appointments

- **Primary Analyzer, Askaryan Radio Array (ARA) Station 1 2021–Present**  
Conducting data analysis of ARA Station 1 for ultra-high energy (UHE) neutrino detection with Prof. D. Besson at University of Kansas (KU).
- **Analysis Lead, Radio Background Characterization with ARA Dec 2025–Present**  
Leading the analysis of astrophysical, environmental, and anthropogenic radio backgrounds in ARA data with Prof. D. Besson at KU.
- **IceCube Neutrino Observatory Monitoring 2021–Present**  
Served as KU representative for monitoring shifts, presenting weekly reports to the collaboration.
- **Machine Learning Project 2025**  
Developed a CNN model for “Automatic Surface Detection in Echograms using Deep Learning” as a part of Machine Learning class at EECS Department, KU, (Project Repository), (Report).
- **Visiting Summer Student Program May–Jul 2019**  
Studied the role of  $\theta_{13}$  mixing angle in three-flavor neutrino oscillations with Prof. S.K. Agarwalla at Institute of Physics, India, producing technical report “Importance of 1-3 Mixing Angle in Three-Flavor Neutrino Oscillations Paradigm” (DOI: 10.13140/RG.2.2.34421.12009).
- **Winter Internship Program Dec 2019–Jan 2020**  
Researched “Weak Interaction Physics” at IIT-BHU with Dr. G. Abbas.

## Field Experience

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### NSF’s Summit Station, Greenland Field Campaign Apr–May 2025

Participated in the Radio Neutrino Observatory – Greenland (RNO-G) calibration campaign, performing in-situ radio-frequency measurements and equipment deployment at Summit Station, Greenland.

## Publications

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### First-Author Works

- **Seikh, M.F.H., Jarvis, R., & Stiles, J. (2026).** “AAFIYA: Antenna Analysis in Frequency-domain for Impedance and Yield Assessment”. arXiv:2601.13583 (Accepted in 2026 IEEE AP-S/URSI Symposium)
- **Seikh, M.F.H. & Besson, D. (2026).** “Polar Ice as a Laboratory for Physics”. *Page 27-37, Physics Bulletin 2026, Physics Department, AMU*
- **Seikh, M.F.H. (2025).** “A geometrical approach to neutrino oscillation parameters”. *AIP Advances 1 October 2025; 15 (10): 105204*. DOI:10.1063/5.0292233.
- **Seikh, M.F.H., Grachev, V., Kravchenko, I., Athar, M.S., & Besson, D. 2025.** “First Radio Neutrino & Cosmic Rays Astronomy Workshop in India”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2509.20704 DOI:10.1140/epjs/s11734-025-01955-8.
- **Seikh, M.F.H., Giri, P. & Besson, D. for the ARA Collaboration 2025.** “Techniques for Continuous Wave Identification and Filtering in the Askaryan Radio Array”. *PoS(ICRC2025)1171*. arXiv:2509.20735.
- **Seikh, M.F.H. & Besson, D. for the RNO-G Collaboration 2025.** “Observation of Radio Solar Flares in RNO-G”. *PoS(ICRC2025)1365*.
- **Seikh, M.F.H. (2024).** “Physics of radio antennas”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.07507 DOI: 10.1140/epjs/s11734-025-01500-7
- **Seikh, M.F.H. for the ARA Collaboration (2024).** “Askaryan Radio Array: searching for the highest energy neutrinos”. *European Physics Journal-Special Topics (EPJ-ST)*. arXiv:2411.01761 DOI: 10.1140/epjs/s11734-025-01708-7

- **Seikh, M.F.H. & Besson, D.** (2024). “IceCube Neutrino Observatory: A Cold Discovery Hub”. *Page 14-23, Physics Bulletin 2024, Physics Department, AMU*
- **Seikh, M.F.H. for the ARA Collaboration** (2023). “Calibration and Physics with ARA Station 1”. *PoS(ICRC2023)1163*. arXiv:2308.07292
- **Seikh, M.F.H.** (2020). “Neutrinos and the 40 Future Routes”. *Physics Bulletin*, Physics Department, AMU. DOI: 10.13140/RG.2.2.27981.10729

## Collaborative Works

Full publication (includes contributions to ARA, RNO-G, ARIANNA, IceCube, IceCube Gen2, PUEO, TAROGE-M and RET collaborations) record available on:

Inspire HEP ; Google Scholar ; ResearchGate ; ORCID  ; Scopus

## Professional Service

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### Editorial

- **Book Editor** for Springer’s EPJ-ST journal: “Radio Detection of Ultra-high Energy Neutrino & Cosmic Ray” (15 articles) with M.S. Athar, D. Besson, V. Grachev, and I. Karvchenko (2024–2025)
- **Founder and Editor-in-Chief** of *The Magnetic Monopole* e-magazine for KU’s Society of Physics Students [View Issue] (June 2022)

### Research Article Review

- **Peer Reviewer:** for World Scientific’s IJMP-A journal (1 article, 2 reviews) (2025)
- **Peer Reviewer:** for Springer’s EPJ-ST journal (5 articles, 11 reviews) (2024–2025)
- **Internal ARA Collaboration Reviewer:** Pawan Giri for ARA Collaboration, “A Comprehensive Diffuse Neutrino Search Using the Full Askaryan Radio Array”, 32nd International Symposium on Lepton Photon Interactions at High Energies (November 2025)

### Conference Session Chair

- **General Cosmology:** for APS Global Physics Summit (March 19, 2026)
- **Topical Groups Session II:** for APS Global Physics Summit (March 17, 2026)

### Other Reviews

- **REU Application Reviewer:** for KU Physics and Astronomy (10 Applications) (2025)
- **SSGSA Application Reviewer:** for SSGSA (Multiple Applications) (2021–2026)

### Professional Leadership/Membership

- **American Physical Society (APS):** Graduate Student Member (2020–Present)
- **National Society of Black Physicists:** Student Member (2021–Present)
- **Physics & Astronomy Graduate Student Organization (GSO), KU:**
  - ✓ Graduate Students Representative - Member at Large (2025–Present)
  - ✓ Graduate Students Representative - Liaison (2024–2025)
  - ✓ GSO Executive Officer (2023–2024)
- **Sir Syed Global Scholar Award (SSGSA):**
  - ✓ Peer Mentor (2023–Present)

- ✓ Team Lead, Promotions & Outreach (2023–2025)
- ✓ Member, Mentorship & Applications Team (2021–2023)
- ✓ AMU Campus Team Member (2019–2020)
- **Society of Physics Students, KU Chapter:** Graduate President (2021–2022)
- **Sigma Pi Sigma (National Physics Honor Society):** Lifetime Member
- **AMU Society of Physics:** Member (2018–Present)

## Technical Proficiencies

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- **Programming Languages:** C/C++, Python, FORTRAN
- **Physics Software:** CERN ROOT, AraRoot, Wolfram Mathematica
- **Simulation Packages:** AraSim, NuRadioMC, HFSS
- **Operating Systems:** Linux (Ubuntu), Windows, macOS
- **Document Preparation:** L<sup>A</sup>T<sub>E</sub>X, Microsoft Office Suite

## Honors & Awards

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### Academic Honors

- **University Gold Medals:**
  - ✓ M.Sc. Physics (Top among 50 students), AMU (2020)
  - ✓ B.Sc. Honors Physics (Top among 120 students), AMU (2018)
  - ✓ Highest percentage in B.Sc. (Hons) Faculty of Science, AMU (2018)
- **Abdul Aziz Medal** for B.Sc. Physics, AMU (2018)
- **Undergraduate Topper Scholarship**, UGC India (2020)
- **University Merit Financial Award**, AMU (2019–2020)
- **First Position** in Higher Secondary (12th) among 40 students (2014)
- **Certificate of Merit** for perfect 10/10 CGPA in Secondary (10th) (2012)

### Fellowships & Grants

- **Ralston Dissertation Fellowship**, KU Physics (2025–2026)
- **McKeithan Summer Fellowship**, KU Physics (2025)
- **Graduate Student Award for Distinguished Service**, KU (2024–2025)
- **Wallace S. Strobel Scholarship**, KU EECS (2024–2025)
- **Ralston Summer Fellowship**, KU Physics (2024)
- **Doctoral Student Research Fund**, KU (2024)
- **Distinguished Students Award**, APS FIP (2021–2022)
- **Carl & Catherine Chaffee Scholarship**, KU Physics (2020–2021)

### Conference Travel Awards

- **Graduate Student Travel Fund**, KU Physics (2023, 2025)
- **DAP Travel Grant**, APS (2024)
- **DPF Travel Award**, APS (2022, 2024)
- **CLAS Graduate Scholarly Development Fund**, KU (2022)

### Other Distinctions

- **Washington DC Scholarship**, AMU Alumni Association (2015–2019)

- **Sir Syed Global Scholar Award** for prospective graduate students (2018)

## Invited Lectures, Seminars, & Colloquia

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- **Radio Engineering for Particle Astrophysics Experiments**  
HEP Seminar, University of Nebraska-Lincoln (March 26, 2025)
- **Radio Engineering for Particle Astrophysics Experiments**  
Astroparticle Seminar, Ohio State University (January 22, 2025)
- **Radio Engineering & Antenna Physics**  
Lecture Series, IEI Summer School 2024, University of Alaska, Anchorage (July 2–5, 2024)
- **UHE Neutrino & Cosmic Ray Detectors**  
ICEHAP Seminar, Chiba University, Japan (August 9, 2023)
- **Antenna Physics & Measurements**  
Lecture Series, IEI Summer School 2023, South Dakota School of Mines (July 2–5, 2023)
- **UHE Neutrino & Cosmic Ray Detectors**  
Physics Colloquium, AMU, India (January 12, 2023)
- **Antennas in UHE Neutrino Detectors**  
Particle Physics Seminar, University of Delaware (October 4, 2022)

## Outreach & Leadership

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- Co-organized and hosted India's inaugural workshop on **Radio Detection of UHE Neutrino and Cosmic Rays** at AMU (April 2024)
- Conducted outreach seminar on **Physics Research Opportunities** at AMU (2021)
- Participated in India's first **Science Leadership Workshop** (2020)
- Volunteer for **One Person One Tree** plantation drive (2017)
- Participant in **Blood Donation Camp** (2017)

## Conference Presentations

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### Oral Presentations

- **Pioneering an Array-Wide Search for Ultra-high Energy Neutrinos with ARA**  
APS Global Physics Summit (March 18, 2026)
- **RNO-G in-ice Antennas**  
RNO-G Collaboration Meeting, Uppsala University (September 9–14, 2024)
- **Radio Frequency Instrumentation**  
Radio Neutrino Workshop, AMU, India (April 18–20, 2024)
- **Towards Five Station ARA Analysis**  
APS April Meeting, Sacramento (April 3–6, 2024)
- **ARA Antenna Models**  
ARA Collaboration Meeting, Ohio State University (March 5–8, 2024)
- **Current Status & Future of ARA**  
APS Prairie Section Meeting, University of Missouri (November 30–December 2, 2023)
- **Introduction to Antenna Physics**  
RET Collaboration Meeting, Vrije University Brussels (November 8–11, 2023)
- **Gain Calibration of RNO-G Antennas**  
RNO-G Fall Meeting, Penn State University (September 11–15, 2023)

- **Galactic Noise in Radio Neutrino Experiments**  
Astro Seminar, KU (*October 28, 2022*)
- **Towards ARA Station 1 Calibration**  
APS April Meeting, New York City (*April 9–12, 2022*)
- **RNO-G Ice Calibration Planning**  
RNO-G Collaboration Meeting, University of Chicago (*April 8, 2022*)

## Poster Presentations

- **A Geometrical Approach to Neutrino Oscillation Parameters**  
APS Global Physics Summit (*March 17, 2026*)
- **CW Identification & Filtering Techniques in ARA**  
39th ICRC, Geneva, Switzerland (*July 19, 2025*)
- **Observation of radio solar flares in RNO-G**  
39th ICRC, Geneva, Switzerland (*July 15, 2025*)
- **Gain Measurements of LPDA Antennas**  
APS April Meeting, Sacramento (*April 3–6, 2024*)
- **Overview of ARA Detector: Five Station Analysis**  
ICTS Neutrino School, India (*April 22–May 3, 2024*)
- **Calibration of ARA Station 1**  
38th ICRC, Nagoya University, Japan (*July 29, 2023*)

## Professional Development

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- **The AI Masterclass 3.0**, (Dhruv Rathee Academy, *May 2026*)
- **Responsible Conduct of Research**, (CITI Program, *2021, 2025*) - 100%
- **From the Big Bang to Dark Energy** (U Tokyo Coursera, *2020*) - 100%
- **Particle Physics: Introduction** (U Geneva Coursera, *2019*) - 100%
- **Senior Diploma in Painting** (Bangiya Sangeet Parishad) - Distinction

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# EPJ ST



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## Special Topics

### Radio Detection of Ultra-high Energy Neutrino and Cosmic Ray

Mohammad Ful Hossain Seikh, Mohammad Sajjad Athar,  
Dave Besson, Victor Grachev, and Ilya Kravchenko (Eds.)



Cosmic rays and neutrinos: exploring high-energy messengers through radio detection.  
Courtesy of Mohammad Ful Hossain Seikh

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